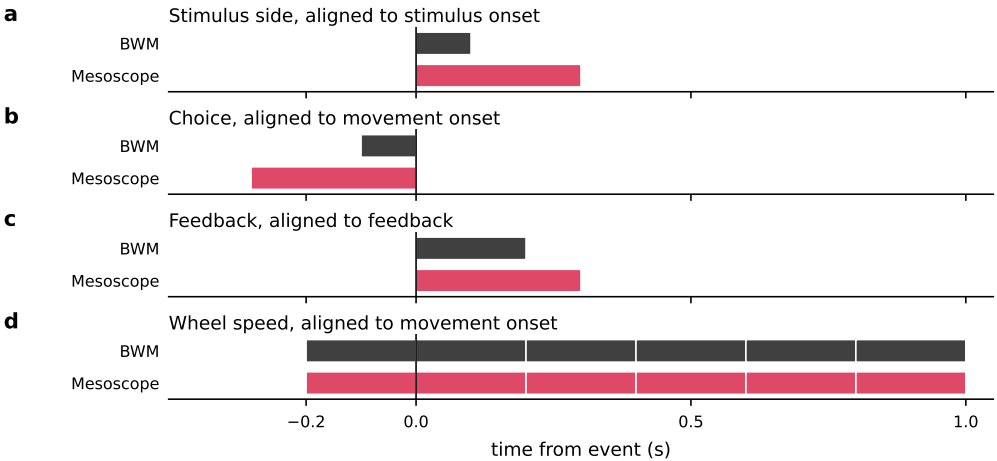


**Figure 1:** Analysis windows used for decoding and for the single-neuron tuning test. BWM windows follow the Brain Wide Map single-cell analyses. Mesoscope frames are about 0.2 s apart, so the mesoscope stimulus, choice and feedback windows are widened to 0.3 s to always contain at least one frame; the mesoscope value is the mean of the deconvolved trace, linearly interpolated at each neuron's own sampling times, over the window. BWM values are firing rates (spike count / window length). **a** Stimulus side: activity in window $[0, 0.1]$ s (BWM) or $[0, 0.3]$ s (mesoscope) relative to stimulus onset. **b** Choice: window $[-0.1, 0]$ s (BWM) or $[-0.3, 0]$ s (mesoscope) relative to movement onset. **c** Feedback: window $[0, 0.2]$ s (BWM) or $[0, 0.3]$ s (mesoscope) relative to feedback. **d** Wheel speed: six 0.2 s bins from $-0.2$ to $1$ s around movement onset, identical for both modalities; each bin's activity is paired with the mean absolute wheel velocity in the same bin.